



REQUESTS: Technical Reference



Project at a Glance

- **Core Modules:** 21
- **Documented Classes:** 70
- **Key Functions:** 81

Table of Contents

- [Ch 1: The Gateway \(API & Core\)](#)
- [Ch 2: The Skeleton \(Data Models\)](#)
- [Ch 3: The Toolbelt \(Utilities\)](#)
- [Ch 4: Supporting Infrastructure](#)

Ch 1: The Gateway (API & Core)

Chapter Architecture

```
graph LR
    sessions --> Imp_0[Impact Score: 0]
    api --> Imp_0[Impact Score: 0]
```

 Module: [src/requests/sessions.py](#)

Classes

- **SessionRedirectMixin**: *No description available.*
- **Session**: A Requests session. Provides cookie persistence, connection-pooling, and configuration. Basic Usage: `>>> import requests >>> s = requests.Session() >>> s.get('https://httpbin.org/get') <Response [200]>` Or as a context manager: `>>> with requests.Session() as s: ... s.get('https://httpbin.org/get') <R...`

Logic

- **merge_setting**: Determines appropriate setting for a given request, taking into account the explicit setting on that request, and the setting in the session. If a setting is a dictionary, they will be merged together using `dict_class`
- **merge_hooks**: Properly merges both requests and session hooks. This is necessary because when `request_hooks == {'response': []}`, the merge breaks Session hooks entirely.
- **session**: Returns a **Session** for context-management. .. deprecated: 1.0.0 This method has been deprecated since version 1.0.0 and is only kept for backwards compatibility. New code should use **requests.sessions.Session** to create a session. This may be removed at a future date. :rtype: Session

 Module: [src/requests/api.py](#)

Logic

- **request**: Constructs and sends a **Request**. method for the new **Request** object: **GET**, **OPTIONS**, **HEAD**, **POST**, **PUT**, **PATCH**, or **DELETE**. URL for the new **Request** object. (optional) Dictionary, list of tuples or bytes to send in the query string for the **Request**. (optional) Dicti...
- **get**: Sends a GET request. URL for the new **Request** object. (optional) Dictionary, list of tuples or bytes to send in the query string for the **Request**. Optional arguments that **request** takes. :return: **Response** object :rtype: requests.Response
- **options**: Sends an OPTIONS request. URL for the new **Request** object. Optional arguments that **request** takes. :return: **Response** object :rtype: requests.Response

Includes: [head](#), [post](#), [put](#), [patch](#), [delete](#)...

Ch 2: The Skeleton (Data Models)

Chapter Architecture

```
graph LR
  models --> Imp_0[Impact Score: 0]
  structures --> Imp_0[Impact Score: 0]
```

 Module: [src/requests/models.py](#)

Classes

- **RequestEncodingMixin**: *No description available.*
- **RequestHooksMixin**: *No description available.*
- **Request**: A user-created **Request** object. Used to prepare a **PreparedRequest**, which is sent to the server. HTTP method to use. URL to send. dictionary of headers to send. dictionary of {filename: fileobject} files to multipart upload. the body to attach to the request. If a dictionary or list of tuples ...
- **PreparedRequest**: The fully mutable **PreparedRequest** object, containing the exact bytes that will be sent to the server. Instances are generated from a **Request** object, and should not be instantiated manually; doing so may produce undesirable effects. Usage: `>>> import requests >>> req = requests.Request('GET',...`
- **Response**: The **Response** object, which contains a server's response to an HTTP request.

 Module: [src/requests/structures.py](#)

Classes

- **CaseInsensitiveDict**: A case-insensitive **dict**-like object. Implements all methods and operations of **MutableMapping** as well as **dict**'s **copy**. Also provides **lower_items**. All keys are expected to be strings. The structure remembers the case of the last key to be set, and **iter(instance)**, **keys()**, **items()**...
 - **LookupDict**: Dictionary lookup object.
-

Ch 3: The Toolbelt (Utilities)

Chapter Architecture

```
graph LR
  utils --> Imp_0[Impact Score: 0]
  _internal_utils --> Imp_0[Impact Score: 0]
```

 Module: `src/requests/utils.py`

Logic

- **dict_to_sequence**: Returns an internal sequence dictionary update.
- **super_len**: *No description available.*
- **get_netrc_auth**: Returns the Requests tuple auth for a given url from netrc.

Includes: `guess_filename`, `extract_zipped_paths`, `atomic_open`, `from_key_val_list`, `to_key_val_list`, `parse_list_header`, `parse_dict_header`, `unquote_header_value`, `dict_from_cookiejar`...

 Module: `src/requests/_internal_utils.py`

Logic

- **to_native_string**: Given a string object, regardless of type, returns a representation of that string in the native string type, encoding and decoding where necessary. This assumes ASCII unless told otherwise.
 - **unicode_is_ascii**: Determine if unicode string only contains ASCII characters. unicode string to check. Must be unicode and not Python 2 `str`. :rtype: bool
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Ch 4: Supporting Infrastructure

Chapter Architecture

```
graph LR
    help --> Imp_0[Impact Score: 0]
    hooks --> Imp_0[Impact Score: 0]
    packages --> Imp_0[Impact Score: 0]
    status_codes --> Imp_0[Impact Score: 0]
    __init__ --> Imp_0[Impact Score: 0]
```

 Module: [src/requests/help.py](#)

Logic

- **`_implementation`**: Return a dict with the Python implementation and version. Provide both the name and the version of the Python implementation currently running. For example, on CPython 3.10.3 it will return `{'name': 'CPython', 'version': '3.10.3'}`. This function works best on CPython and PyPy: in particular, it prob...
 - **`info`**: Generate information for a bug report.
 - **`main`**: Pretty-print the bug information as JSON.
-

 Module: [src/requests/hooks.py](#)

Logic

- **`default_hooks`**: *No description available.*
 - **`dispatch_hook`**: Dispatches a hook dictionary on a given piece of data.
-

 Module: [src/requests/packages.py](#)

 Module: [src/requests/status_codes.py](#)

Logic

- **`_init`**: *No description available.*
-

 Module: [src/requests/__init__.py](#)

Logic

- **`check_compatibility`**: *No description available.*
 - **`_check_cryptography`**: *No description available.*
-

 Module: `src/requests/__version__.py`

 Module: `setup.py`

 Module: `docs/conf.py`

 Module: `docs/_themes/flask_theme_support.py`

Classes

- **FlaskyStyle**: *No description available.*

 Module: `src/requests/adapters.py`

Classes

- **BaseAdapter**: The Base Transport Adapter
- **HTTPAdapter**: The built-in HTTP Adapter for urllib3. Provides a general-case interface for Requests sessions to contact HTTP and HTTPS urls by implementing the Transport Adapter interface. This class will usually be created by the **Session** class under the covers. The number of urllib3 connection pools to cache...

Logic

- **_urllib3_request_context**: *No description available.*

 Module: `src/requests/auth.py`

Classes

- **AuthBase**: Base class that all auth implementations derive from
- **HTTPBasicAuth**: Attaches HTTP Basic Authentication to the given Request object.
- **HTTPProxyAuth**: Attaches HTTP Proxy Authentication to a given Request object.
- **HTTPDigestAuth**: Attaches HTTP Digest Authentication to the given Request object.

Logic

- **_basic_auth_str**: Returns a Basic Auth string.

 Module: `src/requests/certs.py`

 Module: `src/requests/compat.py`

Logic

- **_resolve_char_detection**: Find supported character detection libraries.
-

 Module: `src/requests/cookies.py`

Classes

- **MockRequest**: Wraps a `requests.Request` to mimic a `urllib2.Request`. The code in `http.cookiejar.CookieJar` expects this interface in order to correctly manage cookie policies, i.e., determine whether a cookie can be set, given the domains of the request and the cookie. The original request object is read-only....
- **MockResponse**: Wraps a `httplib.HTTPMessage` to mimic a `urllib.addinfourl`. ...what? Basically, expose the parsed HTTP headers from the server response the way `http.cookiejar` expects to see them.
- **CookieConflictError**: There are two cookies that meet the criteria specified in the cookie jar. Use `.get` and `.set` and include domain and path args in order to be more specific.
- **RequestsCookieJar**: Compatibility class; is a `http.cookiejar.CookieJar`, but exposes a dict interface. This is the CookieJar we create by default for requests and sessions that don't specify one, since some clients may expect `response.cookies` and `session.cookies` to support dict operations. Requests does not use the dict...

Logic

- **extract_cookies_to_jar**: Extract the cookies from the response into a CookieJar.
`http.cookiejar.CookieJar` (not necessarily a `RequestsCookieJar`) our own `requests.Request` object
`urllib3.HTTPResponse` object
- **get_cookie_header**: Produce an appropriate Cookie header string to be sent with `request`, or None.
:rtype: str
- **remove_cookie_by_name**: Unsets a cookie by name, by default over all domains and paths. Wraps `CookieJar.clear()`, is O(n).

Includes: `_copy_cookie_jar`, `create_cookie`, `morsel_to_cookie`, `cookiejar_from_dict`, `merge_cookies...`

 Module: `src/requests/exceptions.py`

Classes

- **UnrewindableBodyError**: Requests encountered an error when trying to rewind a body.
 - **RequestsWarning**: Base warning for Requests.
 - **FileModeWarning**: A file was opened in text mode, but Requests determined its binary length.
 - **RequestsDependencyWarning**: An imported dependency doesn't match the expected version range.
 - **RequestException**: There was an ambiguous exception that occurred while handling your request.
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